

Solvency Field Theory V12.13

Why Particles Weigh What They Weigh

Narrative companion to the
V12.13 Trace-Variational D-Scale Law

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The mass hierarchy of fermions is determined by the geometry of how energy compounds in a closed orbit, protected by the symmetries that prevent spatial settlement from mixing with phase closure.

Publication boundary. V12.13 derives a conditional D-scale mass hierarchy law and a conditional top-sector closure theorem. It does not derive species assignments, the fine structure constant, interaction rules, or the full top/electron closure. It freezes a formula with zero free parameters that hits the base dynamic range to 0.032% and the full hierarchy candidate to 0.63%.

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1 The elevator ride

SFT V12 was an elevator going down, floor by floor, into what a particle is.

V12.4 through V12.5 asked how forces and quantum mechanics arise from a maintained worldline. V12.6 through V12.8 asked which particles can exist — nine modes from no-overwrite and 3+1 slot capacity. V12.9 asked what each particle weighs — mass as geometric focusing. V12.10 asked how particles form — the formation racetrack. V12.11 assembled the Master Action. V12.12 arrived at the basement: the particle is real energy at c , open as light or closed as mass.

V12.13 is the ride back up. It takes the V12.12 particle and asks: now that we know WHAT a particle is, can we derive the mass hierarchy from the geometry of its existence?

The answer is yes. Conditionally.

2 Why energy compounds when it orbits

A photon carries energy through space at c . Its energy creates a gravitational dip — a tiny curvature of spacetime. But the photon moves on. The dip it created fades behind it. The gravitational effect is spread along an infinite open path. No reinforcement. No accumulation. No mass.

A massive particle is the same energy going in circles. It passes through the same region of spacetime every Compton period. Each pass encounters the gravitational dip left by the previous pass. The dip hasn't faded because settlement is permanent — no-overwrite. So each pass deepens the dip slightly. The next pass encounters a deeper dip and deepens it further.

This is compound interest. The settlement grows exponentially because each cycle builds on the accumulated settlement of all previous cycles. The particle's mass is the equilibrium amplitude where the exponential compounding is exactly balanced by local settlement relaxation.

The effective compounding rate per cycle is the closure load Δ_{close} , and it has two components.

3 Two kinds of closure

The energy orbits at c on a circle of radius $r_{\text{spin}} = \hbar/(2mc)$. Each Compton period, it crosses its own curvature 4π radians worth — two full spatial loops of the spinor. This is the spatial/spinor curvature closure. It was derived in R192M from the ratio of the curvature crossing rate $\Omega_r = c/r_{\text{spin}}$ to the Compton frequency:

$$\frac{\Omega_r}{\nu_C} = 4\pi. \quad (1)$$

Simultaneously, the energy's circulation direction — its charge — completes one full 2π rotation of the frame/gauge phase per Compton period. This is not a bookkeeping label. The circulation direction creates real directional settlement structure: same-direction circulations clash (repulsion), opposite-direction circulations mesh (attraction). The frame/gauge phase is physical support, not decoration.

Together, the total closure load per maintenance cycle is:

$$\Delta_{\text{close}} = 4\pi + 2\pi = 6\pi. \quad (2)$$

The half-action — the effective compounding exponent per spatial loop — is $6\pi/2 = 3\pi$.

4 Five components, two sectors

The settlement reinforcement has five independent components. Three come from spatial support: the three normal dimensions of the settlement geometry. Two come from phase closure: the spinor completion and the frame completion.

These five components do not all contribute to the same functional sector. The three spatial components transform under rotations. The two phase components transform under gauge transformations. Rotations and gauge transformations are independent symmetries acting on incommensurable variables. A mixed component that is part-position and part-phase is physically inadmissible — like adding meters to radians.

The five components therefore coarse-grain into exactly two amplitude channels:

Channel A (WHERE): three spatial support coordinates. Weight 3.

Channel B (HOW): two phase closure completions. Weight 2.

The effective participation number from this 3:2 split is:

$$N_{\text{eff}} = \frac{1}{(3/5)^2 + (2/5)^2} = \frac{25}{13}. \quad (3)$$

The amplitude tax — the reduction in effective compounding from having two unequal channels rather than one coherent channel — is $\frac{1}{2} \ln(25/13)$.

5 The D-scale law

The effective feedback load per spatial loop, after the amplitude-channel tax, is:

$$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln\left(\frac{25}{13}\right) \approx 9.0978. \quad (4)$$

The mass function across the nine compression modes is:

$$M_i = G_i \times \exp(L_{\text{eff}} \times u_i), \quad (5)$$

where $u_i = \ln(\rho_i/\rho_0)/\ln 4$ is the normalized compression coordinate and G_i is the weak-geometry prefactor from the mode's winding pattern.

At the heavy endpoint $\rho = 2$, this gives:

$$D_{\text{scale}} = 17,871.489. \quad (6)$$

The R172 computed target from 40 gates of V12.9 feedback calculation is:

$$D_{\text{target}} = 17,877.209. \quad (7)$$

Agreement: **0.032%**.

Every number in L_{eff} traces to SFT-derived structure:

- 3π from half the 6π closure load per maintenance cycle.
- $25/13$ from the 3:2 sector trace with three spatial support coordinates and two phase closure completions.

No measured masses were used. No anchoring to the electron or top quark. No fitting to D . The comparison to D is diagnostic, not definitional.

6 The top-sector closure

The D-scale law gives the base dynamic range of the hierarchy: approximately $18,000\times$ between the lightest and heaviest modes. The full observed span from electron to top quark is approximately $339,000\times$. The ratio is about 18.97.

This ratio is within 0.63% of $6\pi = 18.85$.

V12.13 states this as a conditional theorem: at D-scale saturation, the least nonzero no-overwrite terminal completion that closes both real support sectors is one untaxed $K_{\text{top}} = 4\pi + 2\pi = 6\pi$ layer.

The combined candidate gives:

$$D_{\text{scale}} \times 6\pi = 336,870. \quad (8)$$

The diagnostic target is 339,000. Agreement: 0.63%.

The top-sector theorem is conditional. Its premises — D-scale saturation, no-overwrite minimality, both-sector terminal closure, untaxed completion — are stated and partially supported but not fully proven from the global action.

7 What V12.13 means

The mass hierarchy of fermions is not a set of arbitrary coupling constants. It is the natural consequence of how energy compounds in a closed orbit.

The compounding rate is set by the closure geometry: 6π per maintenance cycle, split into 4π of spinor curvature crossing and 2π of frame/gauge phase completion.

The effective feedback is reduced by the amplitude-channel structure: three spatial dimensions and two phase closures form two incommensurable sectors with a 3:2 weight split, giving an effective participation number of 25/13.

The result is one formula with zero free parameters that reproduces the base dynamic range of the fermion mass spectrum to 0.032% and the full hierarchy candidate to 0.63%.

Whether this formula is correct — whether nature actually computes the mass hierarchy this way — remains to be tested. The species map is open. The prediction suite is open. The global action proof is incomplete. But the STRUCTURE is derived, not fitted. And the structure hits the numbers.

8 The master equation, revisited

V2 said $H = \dot{S}/I$. Five symbols. Expansion equals write rate over information.

V12.13 says: the write rate $\dot{S}$ contains the entire mass hierarchy. Each particle ticks at $\nu_C = mc^2/h$. The mass m is determined by the self-gravitational compounding of the particle's closed orbit, with effective load $L_{\text{eff}} = 3\pi - \frac{1}{2}\ln(25/13)$ set by the closure geometry and the sector structure of three-dimensional space.

The master equation didn't change. Our understanding of what each symbol contains changed. $H = \dot{S}/I$ was always the whole theory. V12.13 is what we now know $\dot{S}$ means.

Acknowledgments

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References

- [1] S. Fosmark, *SFT V12.12: Locally Real Particle Mechanics*, Zenodo (2026).
- [2] S. Fosmark, *SFT V12.12A: Photon Closure-Fit Addendum*, Zenodo (2026).
- [3] S. Fosmark, *SFT V12.9: Mass as Geometric Focusing*, Zenodo (2026).